

Cross-Modality Anatomy Knowledge Distillation for Retinal OCT Layer Segmentation With Limited Labels

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Abstract—Automatic segmentation of retinal Optical Coherence Tomography (OCT) images plays a vital role in the early diagnosis and treatment of ocular diseases. However, the scarcity of labeled OCT data poses a significant challenge for existing supervised learning methods. In contrast, Color Fundus Photography (CFP) provides clearer visualization of key anatomical regions, such as the optic disc and cup, making it easier to acquire high-quality annotations. To address the issue of insufficient OCT annotations, we propose a Cross-Modality Anatomy Knowledge Distillation (CMAKD) framework. This method leverages a large amount of annotated CFP data to learn cross-modality structural priors related to retinal layer organization and vessel-associated

landmarks, which are then used to regularize the OCT model for retinal layer segmentation. In other words, CFP serves as a source of weak structural priors, providing spatial and contextual guidance for OCT feature learning rather than depth-resolved layer supervision or a direct supervisory signal. To achieve this goal, we design a Domain-invariant Contrastive Learning (DoCL) module to reduce the modality gap between CFP and OCT by aligning their feature distributions. Furthermore, we introduce a Layer-aware Self-ensembling Mean-Teacher (LA-MT) framework to fully exploit unlabeled OCT data and enhance the model's ability to recognize layered boundary structures. Experimental results on three public datasets — Duke DME, MS, and GOALS — demonstrate that CMAKD significantly improves segmentation performance while reducing the reliance on OCT annotations. Under semi-supervised domain adaptation settings, CMAKD achieves relative performance gains of 17.8% and 8.2% on the Duke DME and MS datasets, respectively, and approaches the performance of fully supervised models on the GOALS dataset. Our code is available at: <https://github.com/lixiangcog/CMAKD>

Index Terms—Medical image, cross-modality, layer anatomy, retinal OCT segmentation, domain adaptation.

I. INTRODUCTION

OPTICAL Coherence Tomography (OCT), celebrated for its non-radiative and non-invasive characteristics, provides high-resolution cross-sectional images with exceptional detection sensitivity, making it an indispensable tool for the diagnosis and management of a wide range of ocular diseases [1]. For example, the thinning of the Retinal Nerve Fiber Layer (RNFL) is a well-established biomarker for glaucoma, a chronic neurodegenerative disease characterized by the progressive degeneration of retinal ganglion cells and their axons. This degeneration leads to structural damage to the optic nerve head, RNFL thinning, and eventually, irreversible visual field loss [2].

Unlike Color Fundus Photography (CFP), which only captures surface-level retinal information, OCT reveals the retina's cross-sectional, multi-layered structure with fine anatomical detail. This enables clinicians to observe subtle changes in the thickness of critical layers, such as the RNFL and the Ganglion Cell–Inner Plexiform Layer (GCIPL), both of which play a vital role in the early detection and continuous monitoring of ocular diseases. Consequently, OCT provides a more comprehensive assessment of retinal health compared to CFP alone and greatly facilitates the early diagnosis of ocular pathologies [2].

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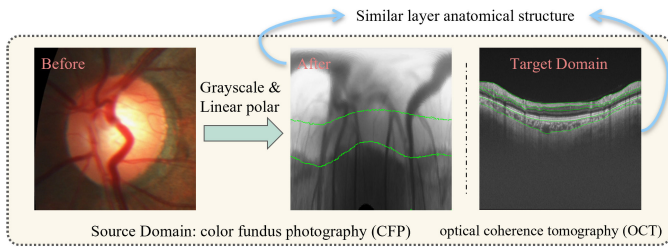


Fig. 1. **Illustration of shared structural cues between the source domain (CFP) and the target domain (OCT).** Despite differences in imaging modalities, biologically related structural priors may support cross-modality feature regularization, especially for superficial retinal structures.

With the rapid advancement of artificial intelligence, deep learning-based methods have been widely explored for automatic OCT retinal layer segmentation. For instance, Wei and Peng [3] introduced a novel fully convolutional network utilizing deep max-pooling to improve segmentation accuracy. Similarly, Cao et al. [4] proposed a multitask learning framework that jointly performs pixel-wise classification and distance map regression to ensure topologically consistent segmentation. More recently, transformer-based architectures [5], [6] have shown promising results in fully supervised OCT segmentation due to their ability to capture long-range dependencies and global context. For instance, Tan et al. [7] proposed a hybrid CNN-Transformer framework (TCCT-BP) that incorporates boundary regression and feature polarization to specifically address inter-layer boundary deviations. However, these methods rely heavily on large-scale, pixel-level annotations, which are challenging and labor-intensive to obtain in clinical practice. To mitigate the reliance on extensive annotations, recent studies have explored adversarial source generation for source-free domain adaptation [8]. However, focusing solely on the target domain may overlook valuable anatomical priors available from other modalities. Although acquiring high-quality OCT annotations is difficult, CFP datasets with simpler segmentation tasks are more readily available. For instance, datasets such as ADAM [9] and REFUGE2 [10] provide optic disc and optic cup annotations for CFP images. The relatively higher contrast and clarity of CFP images in these anatomical regions make accurate annotation more feasible and efficient. These datasets can therefore serve as source domains to assist in training models for more complex target domains, such as OCT segmentation tasks [11], [12]. However, directly transferring knowledge between CFP and OCT is highly challenging due to their inherent differences in imaging modalities, including variations in appearance, anatomical representation, and pixel intensity distribution. While CFP depicts surface-level retinal structures, OCT captures detailed cross-sectional layered anatomy. This modality gap limits the effectiveness of conventional transfer learning approaches.

To overcome these challenges, we propose a novel Cross-Modality Anatomy Knowledge Distillation (CMAKD) framework. Rather than directly transferring labels, our method leverages abundant annotated CFP data to learn shared structural priors across modalities. Clinically, the Optic Disc and Cup (visible in CFP) are biologically associated with the

Retinal Nerve Fiber Layer (RNFL) and Ganglion Cell-Inner Plexiform Layer (GCIPL) in OCT, as RNFL axons converge at the optic nerve head. Additionally, retinal vessel trajectories seen in CFP may correspond to vessel-associated shadows or luminal landmarks in OCT. Our method utilizes these shared structural cues as weak spatial priors to regularize OCT feature learning, rather than assuming direct layer-wise anatomical correspondence between CFP and OCT. This guidance is particularly relevant for superficial retinal structures near the optic nerve head even when OCT annotations are scarce. Importantly, these cues do not directly encode depth-resolved layer identity and are not sufficient to determine retinal layer boundaries on their own; instead, they provide spatial and contextual constraints that narrow the feasible solution space for OCT layer segmentation. The conceptual overview of our approach is illustrated in Fig. 1.

Despite its potential, cross-modality knowledge transfer from CFP to OCT introduces three major technical challenges. First, there is a significant domain discrepancy between CFP and OCT, making it difficult to align their feature distributions effectively. Second, fully exploiting the wealth of unlabeled OCT data in the target domain remains a non-trivial problem, particularly under semi-supervised settings. Third, OCT layer segmentation requires precise boundary delineation of subtle and often nearly linear retinal layers, which adds additional complexity to feature learning and model optimization. To solve these problems, we propose a combination of Domain-invariant Contrastive Learning (DoCL) and a Layer-aware Self-ensembling Mean-teacher (LA-MT) framework. The main contributions of our work are as follows:

- We propose a novel cross-modality anatomy knowledge distillation framework, which is the first to leverage annotated CFP data to provide shared structural priors for OCT retinal layer segmentation, thus alleviating the problem of limited OCT annotations.
- We design the DoCL module, which explicitly reduces the domain discrepancy between CFP and OCT by learning domain-invariant representations.
- We introduce the LA-MT framework, which effectively exploits unlabeled OCT images and enhances the capture of subtle boundary information for layered retinal structures.
- Extensive experiments on three public datasets (Duke DME, MS, and GOALS) demonstrate that our method achieves substantial improvements over state-of-the-art baselines. Specifically, CMAKD achieves relative performance gains of 17.8% and 8.2% on the Duke DME and MS datasets, respectively, and approaches the performance of fully supervised methods on the GOALS dataset.

The rest of this paper is organized as follows. Section II reviews the related work. Section III describes the proposed layer anatomy adaptation framework in detail. Section IV presents and discusses the experimental results. Section V discusses the applicability of CMAKD. Finally, Section VI concludes the paper and outlines possible directions for future research.

II. RELATED WORK

A. Cross Anatomy Domain Adaptation

Transfer Learning (TL) utilizes source dataset knowledge to benefit a new target dataset [13]. Early TL methods primarily involved fine-tuned pre-trained model parameters using target dataset annotations [14]. For instance, Tajbakhsh et al. [15] showed fine-tuning can transfer knowledge from natural to medical images. However, fine-tuning encounters challenges, and its effectiveness diminishes when significant source-target dataset differences exist. Domain Adaptation (DA) combats performance issues when target data differs from the source distribution [16]. It is categorized by annotation nature. Fully-supervised DA adapts a pre-trained model to the target domain through fine-tuning with full annotations, a common choice [17]. In contrast, weakly or semi-supervised DA adapts with limited annotations from the target domain [18]. For example, Li et al. [19] proposed a dual-teacher method for sparse target domain images.

It's worth noting that learning domain-invariant features is an effective approach to address the domain gap. Methods for learning cross-domain invariant representations using deep networks are divided into two categories. The first category involves learning a common network that encourages the learning of invariant feature representations across different domains [20], with the most typical approach being the use of feature discriminators in adversarial training [21]. Although these methods do not introduce additional model parameters, their effectiveness depends largely on the degree of the domain gap. The other direction involves explicitly modeling domain-specific characteristics through a multi-stream network structure, where different domains are modeled by corresponding sub-networks, at the cost of extra parameters [22].

In addition, most annotation-efficient domain adaptation methods are designed for the same modality or modalities with minor differences, which limits their applicability to cross-anatomy scenarios. Recent advances in cross-anatomy domain adaptation have introduced innovative semi-supervised and domain-specific techniques to improve medical image segmentation. For instance, causality-inspired strategies with target style imitation have been proposed to bridge the domain gap [23]. Zhang et al. [24] proposed SS-CADA, which employs vessel-specific normalization and a self-ensembling technique to effectively utilize minimal annotations for coronary artery segmentation. Gu et al. [12] introduced the CS-CADA method, which is based on Domain-specific Batch Normalization (DSBN) [25]. This method combines contrastive learning with a self-ensembling framework to address domain shifts in anatomical structures. These methods often rely on high-quality source domain annotations and are not suitable for layer anatomical structures. In contrast, our method bridges the gap between datasets of different modalities, adapting the layer anatomy from CFP to OCT.

B. Contrastive Learning

Contrastive learning (CL) leverages a loss function to increase similarity among positive pairs and decrease

similarity among negative pairs [26]. Recently, CL has been applied to domain adaptation (DA). For example, Zhang et al. [27] proposed an adaptive contrastive learning framework for unsupervised DA, and He et al. [28] utilized self-training with contrastive learning to address overfitting in partial domain adaptation while Kang et al. [29] developed a contrastive adaptation network to reduce intra-class domain differences and enlarge inter-class separability.

CL-based methods typically require large batches and long training times, limiting their use in medical imaging with limited annotations. To address this, Yeh et al. [30] introduced weighted decoupled contrastive learning (DCLW), which improves training efficiency and stability. More recently, contrastive learning has also been combined with semi-supervised training to improve segmentation robustness across domains. For example, SegCLR [31] integrates supervised segmentation learning with contrastive objectives to enhance representation generalization across multiple OCT domains and enable domain generalization.

Despite these advances, and recent progress in distilling hierarchical knowledge from multimodal fusion [32], most CL-based DA approaches are designed for single-modality tasks and fail to capture the layered anatomical structures of the retina. Our CMAKD framework addresses this gap by integrating CL with anatomical priors, enabling effective knowledge transfer from CFP to OCT for accurate retinal layer segmentation with limited labeled data.

C. Semi-Supervised Learning

Semi-supervised learning (SSL) has recently gained significant attention for medical image analysis, as it enables models to achieve performance close to fully supervised learning using only a small set of labeled data combined with a large amount of unlabeled data [33]. Among SSL techniques, consistency regularization has become a dominant approach. Notably, cooperative teacher-student networks (InterTeach) have been introduced to facilitate semi-supervised medical image segmentation [34]. Yu et al. [35], building on Tarvainen and Valpola's work [36], proposed an uncertainty-aware mean teacher framework that encourages consistent predictions for perturbed versions of the same image, effectively improving segmentation robustness.

Beyond pixel-level consistency, recent research has extended this idea to cross-task consistency, improving generalization across related tasks (Luo et al. [37]; Zhang and Zhang [38]). However, for retinal OCT layer segmentation, SSL remains relatively underexplored [39], [40] due to the unique challenges of OCT images, such as blurred layer boundaries, low contrast, and complex multi-layer structures. To address these difficulties, Fazekas et al. [41] introduced a method leveraging anatomical priors to utilize large-scale unlabeled datasets, while Lu et al. [42] proposed a boundary-enhanced semi-supervised network, representing one of the most advanced OCT segmentation methods to date. Nevertheless, these approaches focus solely on OCT data and do not consider cross-modality information, which can provide rich structural cues for improving segmentation.

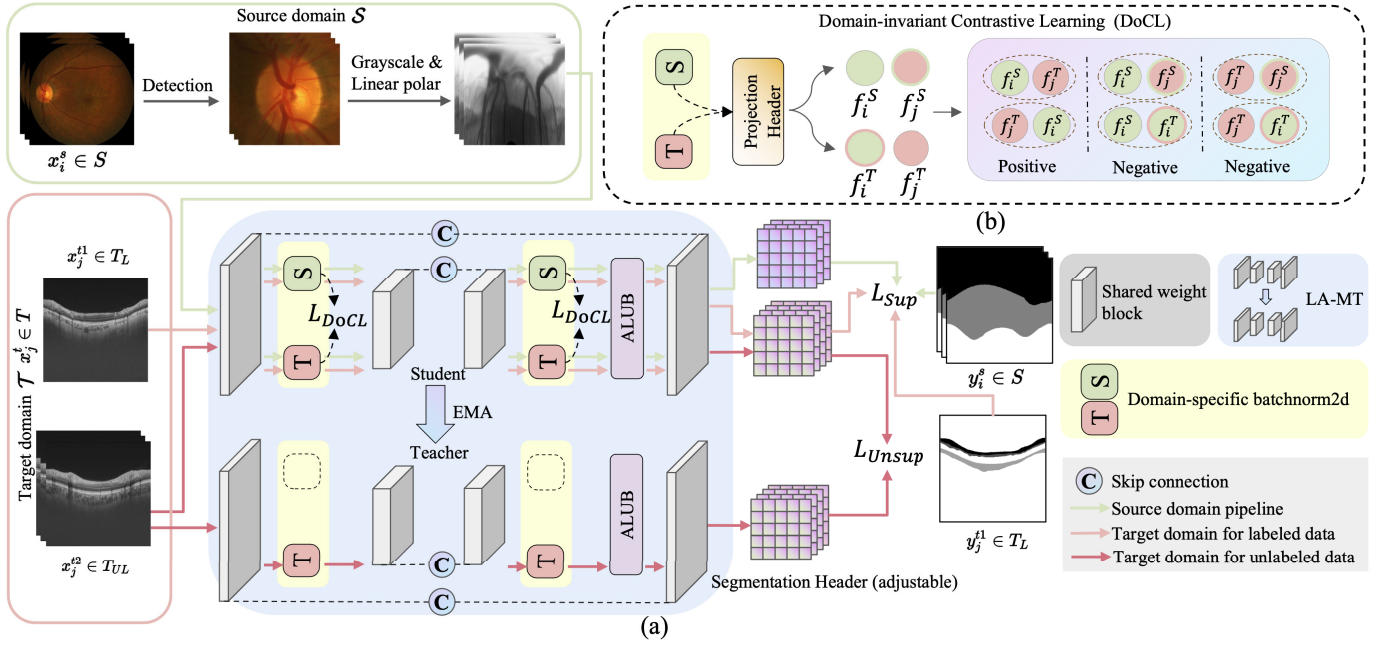


Fig. 2. **The framework of the proposed CMAKD.** f_i^S and f_j^T represent the features extracted from the source-domain image x_i^s and processed by the source-specific and target-specific BatchNorm2d layers of DoCL, respectively. Similarly, f_j^S and f_i^T denote the features extracted from the target-domain image x_j^t and processed by the source-specific and target-specific BatchNorm2d layers of DoCL.

III. PROPOSED METHOD

This section introduces our OCT layer segmentation framework, CMAKD, which comprises three main components (L_{DoCL} , L_{Sup} and L_{Unsup}) as shown in Fig. 2. Initially, the architecture includes a segmentation network (U-Net) [43] consisting of an encoder and a decoder. Incorporating DoCL, the framework extracts features from the source domain (S) and the labeled target domain (T_L), effectively reducing the differences in the invariant feature space between these two domains. In a clinical context, this means the model learns to ignore modality-specific ‘styles’ (such as color in CFP vs. grayscale intensity in OCT) and instead focuses on cross-modality structural cues. Specifically, it aligns representations related to the Optic Disc/Cup regions and vascular patterns (e.g., vessel-associated shadows in OCT corresponding to vessels in CFP), encouraging the feature distribution to capture biologically related structural organization rather than modality-specific appearance. We do not assume direct layer-wise correspondence between CFP and OCT; instead, these cues serve as weak structural priors that regularize OCT feature learning. These priors are not sufficient to determine retinal layer boundaries on their own, but they help bias the model toward anatomically plausible boundary organization when combined with limited labeled OCT supervision. Second, the LA-MT is introduced, which imposes unsupervised consistency on the target domain without labels ($T_{U'L}$), further enhancing data efficiency while improving the perception of layer structures. As depicted in Fig. 2, we denote the existing annotated source domain dataset as $S = \{(x_i^s, y_i^s)\}_{i=0}^{N_s-1}$, and use $T_L = \{(x_j^t, y_j^t)\}_{j=0}^{N_{t1}-1}$ to represent a small set of annotated images in the target domain. Additionally, another set

of unannotated images in the target domain is denoted as $T_{U'L} = \{(x_j^t)\}_{j=0}^{N_{t2}-1}$, where N_s , N_{t1} , and N_{t2} are the number of annotated images in the source domain, annotated images in the target domain, and unannotated images in the target domain, respectively.

A. Domain-Invariant Contrastive Learning (DoCL)

Due to the dissimilar pixel intensity distributions in the source domain S and the target domain T , direct training using both S and T is impractical. Such an approach would be misled by the statistical differences between these domains, making it challenging to learn generalized feature representations effectively.

Different from the two categories of methods mentioned in Section II-A for learning domain-invariant features, our proposed DoCL can encourage the learning of domain-invariant features with fewer additional parameters, thus reducing the domain gap between the source and target domains. As depicted in Fig. 2 (b), DoCL separately designs BatchNorm2d (BN) for the source and target domains. By employing a contrastive learning approach, it divides the features output by the two BNs into positive and negative pairs, and then trains the overall network to acquire the ability to learn domain-invariant features. In the DoCL, input feature maps initially pass through a convolutional block using shared weights. This step is then followed by two distinct BN operations, each operating independently [25]. Specifically, each BN operation computes the mini-batch mean (μ_B) and standard deviation (σ_B), as formulated in Eq. (1). Then, as shown in Eq. (2), BN applies a learnable scale (γ) and a shift (β) parameter to

transform the normalized distribution.

$$\mu_B = \frac{1}{m} \sum_{i=1}^m x_i; \quad \sigma_B = \left[\frac{1}{m} \sum_{i=1}^m (x_i - \mu_B)^2 + \epsilon \right]^{0.5}, \quad (1)$$

$$y_i = \gamma \frac{(x_i - \mu_B)}{\sigma_B} + \beta. \quad (2)$$

Based on the previously reviewed two independent BNs, we incorporate a cross-domain contrastive learning component into their output to address pixel intensity and contextual shifts across different domains and modalities. This component encourages the model to capture domain-invariant features of similar anatomical structures while maintaining robustness to variations in image styles.

Specifically, after gaining the output of the two independent BNs, a nonlinear Projection Head is used to obtain a high-level feature representation. For a given source domain image x_i^s , source-specific and target-specific BN produce normalized feature representations, denoted as f_i^S and f_i^T respectively. Similarly, for a target domain image x_j^t , the same process through the BN module provides normalized features represented as f_j^S and f_j^T . The two independent BNs are designed to normalize feature maps in each domain independently. This allows the feature maps from both domains to be aligned into a common feature distribution space. When given a pair of images from the source and target domains, denoted as x_i^s and x_j^t respectively, f_i^S and f_j^T represent the normalized features in the common distribution space based on their specific BN layers. These features should demonstrate similarity, indicating domain invariance, and are considered positive pairs. Conversely, other feature pairs that do not exhibit similarity are labeled as negative pairs, as illustrated in **Fig. 2 (b)**. For example, the pair (f_i^S, f_i^T) is derived from the same feature of a source-domain image x_i^s normalized by two different BNs. Owing to the distinct styles inherent to the two BNs, they are anticipated to be different from each other, and as a result, are considered as negative pairs.

Algorithm 1 Pseudocode of DoCL in a PyTorch-Like Style

Input normalized embeddings f_1, f_2 ; standard deviation $\sigma = 0.5$; temperature $\tau = 0.1$

Output positive loss $L_+(f_1, f_2)$; negative loss $L_-(f_1, f_2)$

Define $\text{mm}(f_1, f_2)$ to compute cosine similarity.

Define $w(f_1, f_2) = 2 - \text{size}(f_1, 0) \cdot \text{softmax}\left(\frac{f_1 \times f_2}{\sigma}\right)$ to compute adaptive weights.

Define $\epsilon = \log(1 \times 10^{-45})$ to ensure stability of the calculations.

for each pair (f_1, f_2) **do**

$s \leftarrow \text{mm}(f_1, f_2, t())$

$L_+(f_1, f_2) \leftarrow -\text{diag}(s)/\tau \times w(f_1, f_2)$

$s_{\text{neg}} \leftarrow \text{concat}(\text{mm}(f_1, f_1.t()), s)/\tau$

$\text{mask}_{\text{neg}} \leftarrow \text{eye}(\text{size}(f_1, 0)).\text{repeat}(2)$

$L_-(f_1, f_2) \leftarrow \text{logsumexp}(s_{\text{neg}} + \text{mask}_{\text{neg}} \times \epsilon)$

if is_positive **then return** $L_+(f_1, f_2)$

else return $L_-(f_1, f_2)$

end for

Assuming that given any pair of normalized embeddings (f_1, f_2) , the base computation for DoCL is as shown in

Algorithm 1, where the calculation of adaptive weights is inspired by DCLW [30]. Based on the return of Algorithm 1, we define the source-domain to target-domain contrast loss as:

$$L_{\text{DoCL}}^{s \rightarrow t} = L_+(f_i^S, f_j^T) + L_-(f_i^S, f_i^T) + L_-(f_i^S, f_j^S). \quad (3)$$

Correspondingly, we define a target-domain to source-domain contrastive loss as:

$$L_{\text{DoCL}}^{t \rightarrow s} = L_+(f_j^T, f_i^S) + L_-(f_j^T, f_j^S) + L_-(f_j^T, f_i^T). \quad (4)$$

Finally, the DoCL loss, which encourages the learning of domain-invariant features and reduces the domain gap, is formulated as:

$$L_{\text{DoCL}} = \frac{1}{6} (L_{\text{DoCL}}^{s \rightarrow t} + L_{\text{DoCL}}^{t \rightarrow s}). \quad (5)$$

B. Supervision-Based Learning

The second part of CMAKD is supervised training with a U-Net as backbone. Given image annotation pairs $(x_i^s, y_i^s) \in \mathcal{S}$ from the source domain $\mathcal{S}$ and $(x_j^t, y_j^t) \in \mathcal{T}_L$ from the target domain $\mathcal{T}_L$, we define a supervised loss function to optimize network parameters jointly:

$$L_{\text{Sup}} = w_1 \sum_{i=1}^{N_s} L_{\text{DF}}(p_i^s, y_i^s) + w_2 \sum_{j=1}^{N_t} L_{\text{DF}}(p_j^t, y_j^t), \quad (6)$$

where p_i^s and p_j^t represent the network prediction outputs of the CFP on the source domain and the OCT images on the target domain, respectively. L_{DF} denotes a hybrid segmentation loss that consists of the Dice loss and Focal loss:

$$L_{\text{Dice}} = 1 - \sum_{k=1}^K \left(\frac{2 \sum_{i=1}^N p_i y_i}{\sum_{i=1}^N p_i^2 + \sum_{i=1}^N y_i^2} \right), \quad (7)$$

$$L_{\text{Focal}} = \sum_{k=1}^K [-\alpha y_i (1 - p_i)^\gamma \log(p_i) - \alpha (1 - y_i) p_i^\gamma \log(1 - p_i)], \quad (8)$$

$$L_{\text{DF}} = L_{\text{Dice}} + L_{\text{Focal}}, \quad (9)$$

where y_i and p_i represent the actual and predicted categories of the i -th pixel, respectively, and α, γ indicate constant coefficients used to adjust category imbalance. The values of α and γ are set with reference to [44]. In addition, K denotes the total number of images of an input batch on $\mathcal{S}$ or $\mathcal{T}_L$. L_{DF} is generic for $\mathcal{S}$ and $\mathcal{T}_L$.

Additionally, as shown in **Fig. 2 (a)**, the CMAKD segmentation header is adjustable in the number of target domain segmentation categories. For example, in our experiments, the GOALS dataset needs to segment three layers of targets (4 classes); the Duke DME and MS datasets only segment two layers of targets (3 classes), consistent with the OC and OD layer segmentation in the source domain. It should be noted that the class imbalance can be solved by adjusting the two parameters w_1 and w_2 , in Eq. (6) during training.

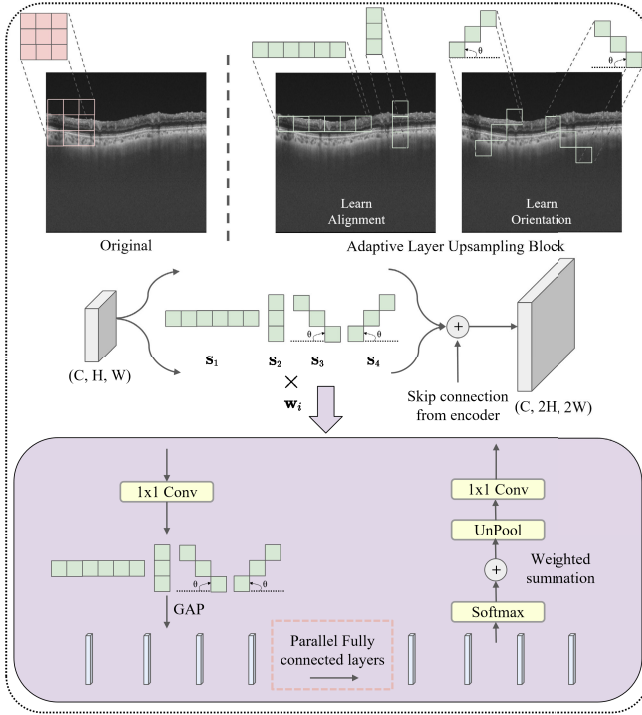


Fig. 3. Adaptive layer Upsampling block (ALUB) in the LA-MT.

C. Layer-Aware Self-Ensembling Mean-Teacher (LA-MT)

To utilize the unannotated OCT images in the target domain, the CMAKD integrates LA-MT framework. It is assumed that the optimization parameters of the student network are $\Theta^{\mathcal{T}}$ and the optimization parameters of the teacher network are $\Theta^{\mathcal{T}'}$. The parameters of the teacher network are computed by the student network, and the parameters of the student network are updated by the gradient descent of the loss function. Specifically, the teacher model $\Theta^{\mathcal{T}'}$ is defined as an Exponential Moving Average (EMA) of the student model in the target domain. In the e -th training step, its parameters can be represented as:

$$\Theta_e^{\mathcal{T}'} = \alpha \Theta_{e-1}^{\mathcal{T}'} + (1 - \alpha) \Theta_e^{\mathcal{T}}, \quad (10)$$

where α is the EMA decay rate.

LA-MT enhances the upsampling layers in the decoders of both the student and teacher networks, optimizing standard decoder upsampling block to Adaptive Layer Upsampling Blocks (ALUB) to better capture the linear features at layer boundaries. Original square convolution kernels commonly used in regular upsampling blocks face difficulties effectively capturing linear features [45], as illustrated in Fig. 3. However, it is worth emphasizing that strip convolution kernels [45] can better capture the linear features at the boundaries of OCT layers, particularly considering the layer anatomical structure.

The ALUB, as illustrated in Fig. 3, is designed to capture contextual information surrounding the CFP or OCT target layers. Unlike previous works such as Strip Pooling [46] and Dynamic Snake Convolution [47], ALUB explicitly models the contextual information surrounding the target layers from horizontal, vertical, and two diagonal directions (with the angle θ being learnable), and achieves adaptive fusion of

multi-directional features through channel-wise attention. ALUB utilizes four strip convolution kernels operating in different directions: horizontal (S_1), vertical (S_2), left diagonal (S_3), and right diagonal (S_4). The ALUB architecture consists of several key components. First, a dimension reduction step is performed using a 1×1 convolutional layer to reduce computational complexity while retaining critical information. Then, strip convolutions capture contextual information from various directions. These convolution outputs undergo global average pooling (GAP) to obtain channel-wise feature vectors. Channel-wise attention is learned using fully connected layers for each strip convolution, enabling the identification of channel-wise importance. Finally, the outputs of the strip convolutions are combined using learned attention weights W_i , resulting in a fused output. The ALUB extracts contextual features for OCT layer segmentation and enables adaptive fusion, enhancing segmentation performance.

Additionally, unlike the self-assembled mean teacher (SE-MT) [36], LA-MT only calculates the loss of consistency between the student and teacher networks in regions with low uncertainty [35]. In particular, we apply the consistency loss L_{Unsup} as the pixel-level mean squared error loss of the teacher and student models only for the most certain predictions:

$$\mathcal{L}_{Unsup}(p^{t2'}, p^{t2}) = \frac{\sum_j \mathbb{I}(u_j < H) \|p_j^{t2'} - p_j^{t2}\|^2}{\sum_j \mathbb{I}(u_j < H)}, \quad (11)$$

where $\mathbb{I}(\cdot)$ is the indicator function; $p_j^{t2'}$ and p_j^{t2} are the predictions of the teacher network and student network at the j -th pixel, respectively; u_j is the estimated uncertainty at the j -th pixel, and H is a threshold to select the most certain targets.

D. Overall Training Loss

The overall loss for training is a combination of the contrastive loss L_{DoCL} from Eq. (5), the supervised loss L_{Sup} from Eq. (6), and the consistency loss L_{Unsup} from Eq. (11). The formulation is as follows:

$$L = \lambda_1 \cdot L_{DoCL} + \lambda_2 \cdot L_{Sup} + \lambda_3 \cdot L_{Unsup}, \quad (12)$$

where λ_1 , λ_2 , and λ_3 act as trade-off parameters. We optimize the network parameters of U-Net jointly using these three loss functions. Once the training process is completed, for the inference process, segmentation results for OCT images are obtained by performing forward propagation on the student model $\Theta^{\mathcal{T}}$.

By seamlessly integrating cross-modality DoCL, supervised learning, and LA-MT, the CMAKD framework enables cross-modality structural prior transfer and feature regularization from CFP to OCT. This approach fully leverages the rich annotated information from CFP and the abundant unlabeled data from OCT, effectively narrowing the gap between the two imaging modalities and significantly enhancing the model's ability to perceive the layered retinal structures in OCT images.

IV. EXPERIMENTS

A. Dataset and Implementation

In the source domain, we used the REFUGE2 dataset, while in the target domain, CMAKD was trained, validated and tested using the GOALS challenge, Duke DME, and MS datasets. Additionally, we used random horizontal and vertical flipping, and random cropping for data augmentation. Notably, these operations shuffle the fixed left–right correspondence, thereby mitigating the bias introduced by anatomical differences between right and left eyes. The image intensity is normalized to $[0, 1]$.

The REFUGE2 dataset [10] is a publicly available CFP dataset focused on glaucoma, with segmentation annotations for the optic Disc (OD) and Optic Cup (OC). Our experiment used 2000 images with OC and OD segmentation labels as the source domain. To extract ROI regions for subsequent polar coordinate transformation operations, we initially performed coarse segmentation of CFP using U-Net to detect OC and OD. Therefore, the sample size of the source domain $|S| = 2000$.

The GOALS dataset [2] is a public circumpapillary OCT dataset related to glaucoma. It provides annotations of layer segmentation masks for the RNFL, GCIPL, and choroid layer. In our experiment, we used 90 labeled images for the training set and 10 images for the validation set. Within the training set of 90 images, we only used a supervised training approach with a ratio of 0.2 labeled images. The original images are sized at 800×1100 , and we resized and center-cropped them into 256×256 patches. Each sample can obtain four patches, resulting in $|T_L| = 72$ and $|T_{U'L}| = 288$. The final test results were evaluated on 100 OCT test images, which correspond to the data from the validation folder in the GOALS competition, essentially serving as the preliminary test set.

The Duke DME dataset [48] consists of 110 macular B-scans from 10 patients with severe DME pathology. The OCT images provide multiple layer segmentation labels, but in our experiment, we focused only on the RNFL and GCIPL layers, as they are closely related to early glaucoma screening and diagnosis [49]. According to the official standards, the dataset is divided into 66, 22, and 22 images for the training, validation, and testing sets. For the Duke DME dataset, the training, validation, and test splits are performed at the patient level to prevent potential information leakage between images from the same subject. Similarly, we only used a ratio of 0.2 labeled images from this dataset for supervised training. The original image size is 496×536 , and we resized and center-cropped them into 256×256 patches. Each sample can obtain two patches, resulting in $|T_L| = 26$ and $|T_{U'L}| = 106$.

The MS dataset [50] comprises macular OCT data from 35 individuals, including 14 healthy controls (HC) and 21 patients with multiple sclerosis (MS). Each subject's OCT data contains 49 B-scans, each sized at 496×1024 . Our experiment only focused on the RNFL and GCIPL layers. We divided each B-scan into four non-overlapping samples in the width direction and cropped these samples to the pixel height. This way, we obtained 6860 retinal OCT scans sized at 256×256 . We divided them into three parts: a training set of 5880 scans from 30 subjects, a validation set of 392 scans from 2 subjects, and

TABLE I

QUANTITATIVE COMPARISON BETWEEN OUR PROPOSED CMAKD AND FULLY SUPERVISED METHODS IN THE GOALS CHALLENGE. IT IS WORTH NOTING THAT IN THE TARGET DOMAIN OF CMAKD, ONLY 20% OF THE TRAINING IMAGES ARE ANNOTATED

Method		DICE $\uparrow$		MED $\downarrow$	
		RNFL	GCIPL	RNFL	GCIPL
Fully Supervised	U-Net (baseline) [2]	0.8161	0.6295	4.0597	3.3180
	TransUNet* [5]	0.7099	0.6516	4.3211	2.9843
	SegFormer* [6]	0.8418	0.7771	3.1820	3.0004
CMAKD (proposed) [†]		0.7593	0.8088	3.2910	2.7992

a testing set of 588 scans from the remaining 3 subjects. We only used 5 annotations (20 patches) from different subjects for training. Hence, $|T_L| = 20$ and $|T_{U'L}| = 5860$.

We implemented CMAKD using torch-1.8.1-cu11.1-cudnn8 and conducted experiments on a workstation equipped with a 6-core CPU and an NVIDIA GeForce RTX 3090 GPU (24 GB VRAM, 1 card). Regarding some hyperparameters, the EMA decay rate was empirically set to 0.98. We used a time-dependent Gaussian warm-up function $\alpha(i) = 0.1 e^{-5(1-i/i_{\max})^2}$ to dynamically adjust the hyperparameter α in Equation (10), where i represents the current training iteration and $i_{\max}$ is the last iteration. The warm-up mechanism was employed to facilitate better network convergence and prevent it from getting stuck in local optima. Every model underwent training using the Adam optimizer, where a maximum of 40k iterations were allowed, and the initial learning rate was set to $1e-4$. The learning rate was subject to exponential decay with a decay factor of 0.95. During training, the loss function weights were set as $\lambda_1 = 0.4$, $\lambda_2 = 1$, and $\lambda_3 = 0.6$ in Eq. (12), respectively. The batch size was 8, with half of the images labeled and the other half unlabeled in each batch.

B. Evaluation Metrics

The DICE coefficient serves as an evaluation metric for assessing the segmentation results [44].

Furthermore, we used the Euclidean distance to measure the error between the segmented result's boundary and the ground truth's boundary. Specifically, we iterate through each pixel point on the segmented result's boundary and calculate the Euclidean distance to the nearest pixel point on the ground truth's boundary. Then, we compute the average of these Euclidean distances based on the number of pixel points on the segmented result's boundary:

$$MED = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_i - x_i^0)^2 + (y_i - y_i^0)^2} \quad (13)$$

where N is the number of pixels on the edge of the segmentation result, (x_i, y_i) represents the i -th pixel on the edge of the segmentation, and (x_i^0, y_i^0) represents the nearest pixel on the gold standard edge to (x_i, y_i) .

In **Tables I, II, and III**, * indicates that the method was implemented based on the open-source code; [†] indicates that the CFP in the source domain underwent grayscale and linear polar coordinate transformations; remaining results are cited from other relevant literature.

TABLE II

COMPARISON OF OUR PROPOSED METHOD, CMAKD, WITH OTHER SOTA METHODS ON THE DUKE DME DATASET. IT IS WORTH NOTING THAT IN THE TARGET DOMAIN OF CMAKD, ONLY 20% OF THE TRAINING IMAGES ARE ANNOTATED. FINE-TUNING (LAST LAYER) SPECIFICALLY INVOLVES UPDATING THE PARAMETERS OF THE LAST CONVOLUTIONAL BLOCK IN THE DECODER. ON THE OTHER HAND, FINE-TUNING (ALL LAYERS) ENTAILS UPDATING ALL PARAMETERS OF THE MODEL

	Method	Training set S T_L T_{UL}	GCIPL_DICE	RNFL_DICE	Mean_DICE
	Inter-observer [59]		0.8900	0.8600	0.8750
Fully Supervised	U-Net	✓	0.9000	0.8500	0.8750
	ReLayNet [60]	✓	0.9200	0.8800	0.9000
SAM-based (zero shot)	SAM-Med2D* [52]		0.6241	0.5892	0.6067
	Medical SAM3* [53]		0.6518	0.5834	0.6176
SAM-FT	SAM-UNet* [54]	✓	0.8872	0.8415	0.8644
	UniOCTSeg* [55]	✓	0.9105	0.8642	0.8874
Semi-supervised	SE-MT [36]	✓ ✓	0.9076	0.8547	0.8812
	UA-MT [35]	✓ ✓	0.9283	0.8680	0.8982
	SD-LayerNet* [41]	✓ ✓	0.9312	0.8611	0.8961
	DiffTCL* [56]	✓ ✓	0.9345	0.8602	0.8974
Supervised DA	Joint Training* [†]	✓ ✓	0.8120	0.7137	0.7629
	Fine-tuning (last layer)* [†]	✓ ✓	0.5416	0.4754	0.5085
	Fine-tuning (all layers)* [†]	✓ ✓	0.8914	0.8839	0.8877
Semi-supervised DA	DSBN* [†] [25]	✓ ✓ ✓	0.8379	0.7150	0.7765
	SS-CADA* [†] [11]	✓ ✓ ✓	0.8343	0.7202	0.7773
	CS-CADA* [†] [12]	✓ ✓ ✓	0.8456	0.7244	0.7850
	CD-BIER* [†] [57]	✓ ✓ ✓	0.8312	0.7188	0.7750
	SEN-MTKD* [†] [58]	✓ ✓ ✓	0.8395	0.7210	0.7803
Semi-supervised DA	CMAKD (proposed) [†]	✓ ✓ ✓	0.9433	0.9055	0.9244

C. Comparison With Fully Supervised Methods on the GOALS

We performed a quantitative comparison with the official baseline on the GOALS dataset. Although the DICE values of the RNFL layer were lower than the baseline, CMAKD performed better on the DICE values and MED values of the other layers, as shown in **Table I**. It can be observed that our method performs roughly on par with other SOTA fully supervised methods. It is worth noting that retinal OCT layer segmentation is a relatively structured task, where strong anatomical regularity allows even simple architectures to achieve high Dice scores, leaving limited room for large numerical improvements. Crucially, while these baselines rely on 100% of the annotated dataset, CMAKD achieves this competitive performance using only 20% of the labeled OCT images. This demonstrates that the OCT annotation workload can be effectively reduced to approximately one-fifth without compromising segmentation quality.

D. Comparison With SOTA Methods on the Duke DME

We compared it with other SOTA methods on the Duke DME test dataset, as shown in **Table II**. The compared methods include expert manual annotation, supervised learning, SAM-based zero-shot, fully supervised fine-tuning of SAM (SAM-FT), semi-supervised learning, supervised DA, and semi-supervised DA methods. The results show that CMAKD achieves superior performance on both GCIPL and RNFL layers in OCT images. Among them, GCIPL_DICE improved by

about 4.8%, and RNFL_DICE improved by about 6.5% over the traditional supervised U-Net. This is because CMAKD can fully utilize the source domain image S , the annotated image T_L in the target domain, and the unannotated image T_{UL} in the target domain simultaneously. Compared to other semi-supervised DA methods with the same settings, such as CS-CADA, the performance relatively increased by 17.8%. In **Fig. 4**, we perform a qualitative comparison of methods, including U-Net, and UA-MT, Fine-tuning (all layers), CS-CADA, all from the test dataset of Duke DME. It can be found that the segmentation results of CMAKD are closer to the gold standard. It is worth noting that our method achieves significant improvements over semi-supervised methods that rely solely on OCT data, outperforming the top-performing UA-MT by approximately 2.6% in Mean DICE. It is worth noting that semi-supervised methods achieve relatively strong performance in **Table II** compared with several domain adaptation baselines. One possible reason is that the source and target domains in our setting correspond to CFP and OCT images, which differ substantially in imaging mechanisms and visual characteristics. Directly introducing CFP as the source domain may therefore introduce representation mismatch or noise for OCT segmentation, making optimization difficult for conventional domain adaptation approaches. In contrast, the proposed CMAKD framework leverages CFP images to extract anatomical priors and integrates them into the OCT segmentation process through dedicated modules, which helps alleviate cross-modality discrepancies and improves the effectiveness of knowledge transfer.

TABLE III

COMPARISON OF OUR PROPOSED METHOD, CMAKD, WITH OTHER SOTA METHODS ON THE MS DATASET. IT IS WORTH NOTING THAT IN THE TARGET DOMAIN OF CMAKD, ONLY 5 TRAINING IMAGES FROM DIVERSE SUBJECTS ARE ANNOTATED

	Method	Training set			GCIPL_DICE	RNFL_DICE	Mean_DICE
		S	T_L	T_{UL}			
Fully Supervised	U-Net	✓			0.8939	0.8412	0.8676
	MGU-Net [51]	✓			0.9019	0.8950	0.8985
	MTF [4]	✓			0.8810	0.8951	0.8881
SAM-based (zero shot)	SAM-Med2D* [52]				0.6542	0.6013	0.6278
	Medical SAM3* [53]				0.6811	0.6354	0.6583
SAM-FT	SAM-UNet* [54]	✓			0.8955	0.8250	0.8603
	UniOCTSeg* [55]	✓			0.9067	0.8873	0.8970
Semi-supervised	SE-MT [36]	✓	✓		0.8992	0.8900	0.8946
	UA-MT [35]	✓	✓		0.9006	0.8913	0.8960
	DTC [37]	✓	✓		0.9044	0.8926	0.8985
	SD-LayerNet* [41]	✓	✓		0.9090	0.8557	0.8824
	BE-SemiNet [42]	✓	✓		0.9112	0.8962	0.8982
	DiffTCL* [56]	✓	✓		0.8865	0.8963	0.8914
Supervised DA	Joint Training* [†]	✓	✓		0.8492	0.8014	0.8253
	Fine-tuning (last layer)* [†]	✓	✓		0.6609	0.5946	0.6278
	Fine-tuning (all layers)* [†]	✓	✓		0.8830	0.8796	0.8813
Semi-supervised DA	DSBN* [†] [25]	✓	✓	✓	0.8445	0.8143	0.8294
	SS-CADA* [†] [11]	✓	✓	✓	0.8371	0.8305	0.8338
	CS-CADA* [†] [12]	✓	✓	✓	0.8523	0.8204	0.8364
	CD-BIER* [†] [57]	✓	✓	✓	0.8410	0.8101	0.8256
	SEN-MTKD* [†] [58]	✓	✓	✓	0.8463	0.8182	0.8323
Semi-supervised DA	CMAKD (proposed) [†]	✓	✓	✓	0.9182	0.8909	0.9046

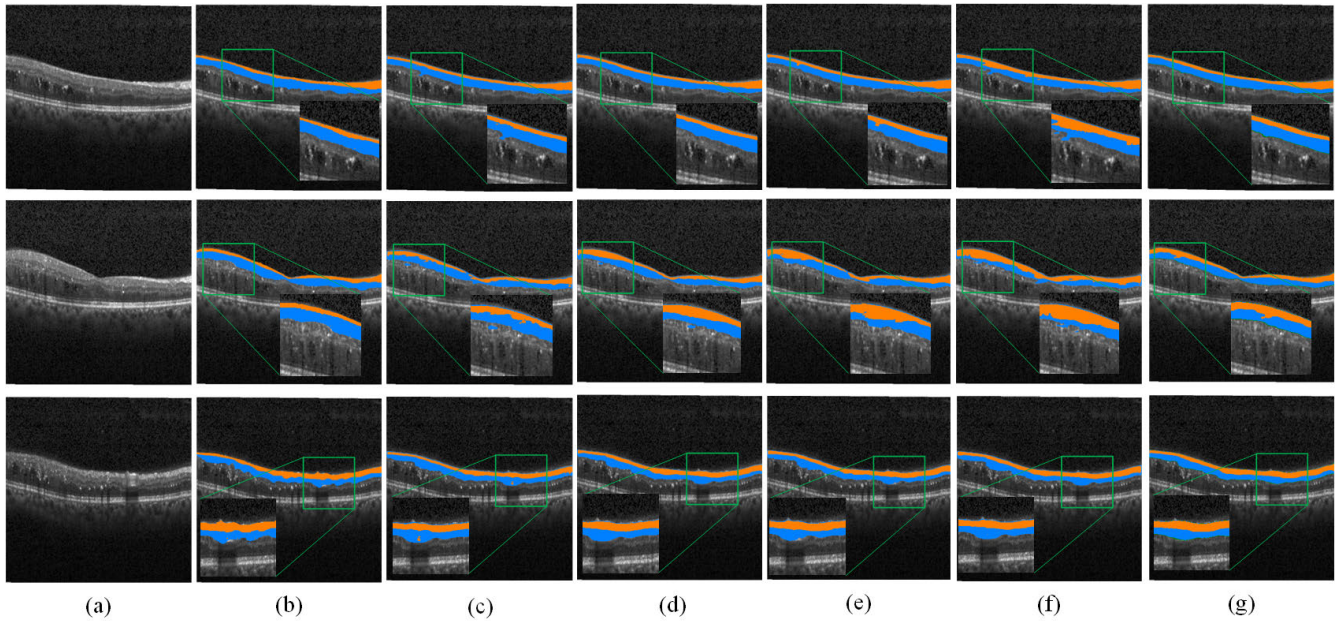


Fig. 4. Segmentation results of different models on the Duke DME test dataset. (a) Original OCT gray images, (b) ground truth, (c) U-Net, (d) UA-MT, (e) fine-tuning (all layers), (f) CS-CADA, and (g) our proposed CMAKD.

E. Comparison With SOTA Methods on the MS

We conducted a comprehensive comparison of our approach with other SOTA methods using the MS dataset, as outlined in Table III. These methods can be divided into six categories: supervised methods (U-Net, MGU-Net [51], and MTF), SAM-based zero-shot methods (SAM-Med2D

[52] and Medical SAM3 [53]), SAM-FT methods (SAM-UNet [54] and UniOCTSeg [55]), semi-supervised methods (SE-MT, UA-MT, BE-SemiNet, and DiffTCL [56]), supervised domain adaptation methods (Joint Training, Fine-tuning, etc.), and semi-supervised domain adaptation methods (DSBN, SS-CADA, CS-CADA, CD-BIER [57], and SEN-MTKD [58]). This structured analysis allows us to assess the relative

TABLE IV

ABLATION EXPERIMENTS ON THE GOALS DATASET. “BASELINE (TARGET)” REPRESENTS THE ABSENCE OF OPTIMIZATION USING LA-MT AND DoCL, AND TRAINING IS CONDUCTED USING ONLY TARGET DOMAIN DATA

Method	Mean_DICE $\uparrow$	Mean_MED $\downarrow$	FLOPs/G $\downarrow$	Params/M $\downarrow$
Baseline (Target)	0.4685	23.994	10.91	6.07
Baseline + LA-MT	0.7006	10.922	19.43	10.74
Baseline + DoCL (using CFP data)	0.7503	8.1263	11.05	6.17
CMAKD w/o ALUB (using MaxPool2d)	0.7819	5.4160	19.02	10.01
CMAKD w/o CFP linear polar transformation	0.5149	15.615	20.93	10.84
Baseline + LA-MT + DoCL (CMAKD)	0.8073	4.0803	20.93	10.84

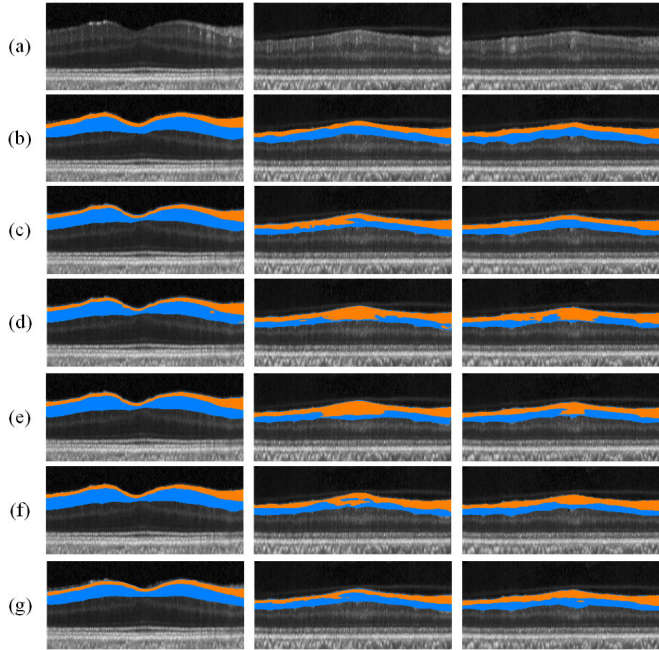


Fig. 5. Segmentation results of different models on the MS test dataset. (a) Original OCT gray images (b) Ground truth, (c) U-Net, (d) UA-MT, (e) Fine-tuning (all layers), (f) CS-CADA, (g) our proposed CMAKD.

performance of our approach and provides a clear overview of the results and metrics. From **Table III**, although the performance of CMAKD on the RNFL layer is slightly lower than the semi-supervised BE-SemiNet method, our approach achieves a higher overall average performance across the evaluated layers. While the performance gap is relatively small, it is worth noting that BE-SemiNet relies on complex pre-processing and post-processing operations for OCT layer boundaries, whereas CMAKD does not require such additional steps. Under this setting, the results indicate that incorporating cross-modality anatomical priors from CFP can still provide useful structural guidance for OCT layer segmentation under limited annotations, suggesting the effectiveness of integrating external anatomical knowledge into semi-supervised OCT segmentation frameworks. CMAKD achieves DICE values of 0.9182 and 0.8909 on the GCIPL and RNFL layers of the OCT images, respectively, and has the highest average DICE value. Compared to other semi-supervised DA methods with the same settings, such as CS-CADA, the performance relatively increased by 8.2%. In **Fig. 5**, we performed a qualitative

comparison of methods including U-Net, and UA-MT, Fine-tuning (all layers), CS-CADA, all from the test dataset of MS. It can be found that the segmentation results of CMAKD are closer to the gold standard and there are fewer over- and under-segmented pixels.

F. Ablation Analysis

In this section, we perform an ablation analysis of the CMAKD, as shown in **Table IV**. It can be found that when three loss optimizations are used simultaneously (L_{Sup} , L_{Unsup} , and L_{DoCL} , corresponding to the last row of **Table IV**), the performance is optimal, with the highest average DICE value of 0.8073 and the lowest average MED of 4.0803. Additionally, if the CFP data in the source domain is not transformed using linear polar coordinates, performance decreases, indicating that transforming to a linear anatomical structure aids in adaptation. Furthermore, as observed from **Table IV**, there is almost no increase in floating point of operations (FLOPs) and the number of parameters [44], which demonstrates that DoCL does not introduce additional computational overhead. While LA-MT increases FLOPs and parameters compared to the Baseline, during inference, we only use the student model Θ^T , allowing the FLOPs and parameter count incurred by training the teacher model to be ignored.

As shown in **Fig. 6**, the qualitative results demonstrate that the proposed method helps improve segmentation accuracy at object boundaries. Compared with CMAKD w/o DoCL and CMAKD w/o ALUB, the full CMAKD produces substantially fewer pixel-wise errors around elongated retinal layer boundaries, which are the most challenging regions in OCT images due to weak inter-layer contrast, speckle noise, and the layered structure. This observation suggests that ALUB is able to enhance directional contextual modeling along retinal layers and their oblique transition regions, while DoCL promotes boundary-discriminative and cross-domain consistent representation learning in ambiguous interface regions.

Additionally, we conducted ablation experiments (**Fig. 7**) to assess the impact of the number of visible annotated target domain images and the patch size on network performance. In the Duke DME dataset (**Fig. 7 (a)**), the network performance significantly improves when the ratio of visible annotated images reaches 0.2, and high performance is achieved with the minimum number of such images. Similarly, in the MS dataset, we conducted ablation experiments (**Fig. 7 (b)**) to explore the influence of the number of visible annotated images and

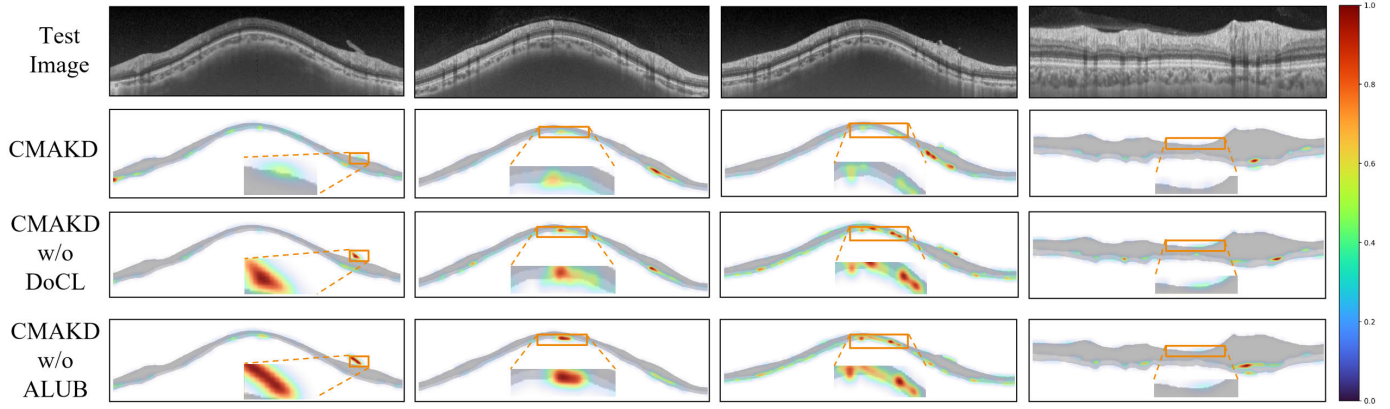


Fig. 6. **Pixel-wise error heatmaps for representative samples from the GOALS dataset.** Compared with CMAKD w/o DoCL and CMAKD w/o ALUB, the full CMAKD exhibits less concentrated error around elongated retinal layer boundaries, indicating that DoCL and ALUB contribute to boundary-sensitive OCT representation learning.

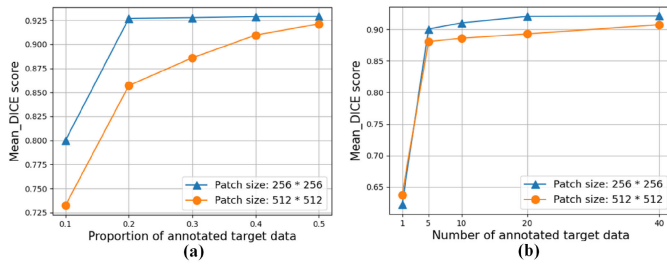


Fig. 7. **Average Dice Coefficient.** (a) under different ratios of annotated OCT images on the Duke DME dataset. (b) under different numbers of annotated OCT images on the MS dataset. The blue and orange lines represent two distinct patch sizes used for image cropping.

TABLE V

PERFORMANCE OF CMAKD AND FULLY SUPERVISED METHODS FOR CHOROIDAL LAYER SEGMENTATION IN THE GOALS CHALLENGE

	Method	DICE $\uparrow$	MED $\downarrow$
Supervised	U-Net (baseline) [2]	0.8193	8.9155
	TransUNet* [5]	0.7460	7.5155
	SegFormer* [6]	0.8449	6.7402
	CMAKD (proposed) [†]	0.8539	6.1509

patch size. When the number of samples is 5 (equivalent to 20 patches), better segmentation results can be obtained with the minimum number of visible annotated images. It can be observed that a patch size of 256 performs better than 512. This is because when the size of the center-cropped OCT image is 256, the pixel intensity distribution is closer to the transformed CFP.

V. DISCUSSION

A. Exploring Choroidal Layer Segmentation

Although the primary clinical focus of this study is on glaucoma-related biomarkers (RNFL and GCIPL), we further evaluated the generalization capability of the CMAKD framework by extending it to choroidal layer segmentation on the GOALS dataset. Unlike the inner retinal layers, the

choroid is located deeper in the posterior part of the eye and presents significant segmentation challenges, specifically lower contrast, complex vascular textures, and ambiguous boundaries compared to the highly reflective RNFL. Despite these anatomical disparities and the fact that the source domain (CFP) models are trained only on Optic Disc/Cup regions, CMAKD effectively transfers generalized structural priors—such as boundary continuity and layer topology—to the choroid task. As shown in Table V, our method achieves a DICE score of 0.8539 and an MED of 6.1509, significantly outperforming state-of-the-art fully supervised baselines like U-Net and TransUNet under limited supervision. These results demonstrate that our framework is not strictly constrained by the specific semantic categories defined in the source domain; instead, it possesses a robust capability to adapt learned anatomical knowledge to distinct ocular structures, affirming its potential for broader applications in diagnosing choroidal diseases such as high myopia.

B. Extension to Cardiac Image Segmentation

In the previous section, CMAKD demonstrated outstanding cross-modal segmentation performance, transitioning from CFP to OCT images, both quantitatively and qualitatively. To further exhibit CMAKD's generalization capability, we extended our experiments to a cross-modal scenario, moving from CFP to CMR images, on the M&Ms dataset [61]. The dataset is a public CMR dataset comprising 375 participants who were scanned using four different scanners. The dataset has been annotated consistently across various centers using standardized contouring. We are only concerned with the regions of the left ventricle (LV) blood pools and the left ventricular myocardium (MYO) in the images. We need to slice the 3D volume of this dataset, and because some cases in the dataset have missing labels, we need to perform filtering. After preprocessing, we obtained 1021 sliced images along with their corresponding labels. We divided them into training, validation, and test sets in a ratio of 3:1:1. In the end, the training set consisted of a total of 613 slices, with $|T_L| = 123$ and $|T_{U'L}| = 490$.

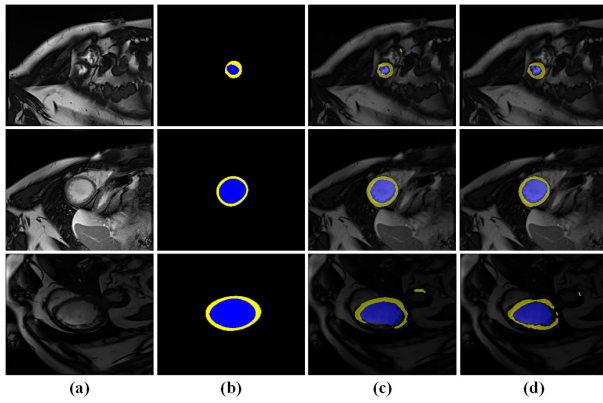


Fig. 8. Segmentation results of different models on the M&Ms test dataset. (a) Original gray images, (b) ground truth, (c) SegFormer, and (d) our proposed CMAKD.

TABLE VI

COMPARED TO THE FULLY SUPERVISED METHOD ON THE M&Ms DATASET, CMAKD REQUIRES ONLY 20% OF THE TRAINING SET WITH LABELED IMAGES. THE Δ IS MEASURED AGAINST SEGFORMER

Method	LV_DICE	MYO_DICE	Mean_DICE
SegFormer (Supervised) [6]	0.8799	0.8584	0.8692
CMAKD (proposed)	0.8874	0.8534	0.8704
Gain (Δ)	$\uparrow 0.0075$	$\downarrow 0.0050$	$\uparrow 0.0012$

TABLE VII

SATURATION ANALYSIS OF INCREASING UNLABELED OCT DATA ON THE DUKE DME DATASET. WE COMPARE THE EFFECT OF INCREASING SAME-MODALITY UNLABELED OCT DATA WITH THAT OF THE PROPOSED CROSS-MODALITY CFP ASSISTANCE

Method	Extra Data	Dice (%) $\uparrow$	Gain Δ
SD-LayerNet	-	89.61	-
SD-LayerNet	+ 100 Unlabeled OCT	89.75	+0.14
SD-LayerNet	+ 500 Unlabeled OCT	89.88	+0.27
SD-LayerNet	+ 1,000 Unlabeled OCT	89.92	+0.31
DiffTCL	-	89.74	-
DiffTCL	+ 100 Unlabeled OCT	89.92	+0.18
DiffTCL	+ 500 Unlabeled OCT	90.11	+0.37
DiffTCL	+ 1,000 Unlabeled OCT	90.19	+0.45
CMAKD	using 1,000 labeled CFP	90.85	-
CMAKD	using 2,000 labeled CFP	92.44	+1.59

Notably, to maintain structural similarity between the optic disc/cup region in CFP and the MYO/LV targets in CMR images, we omitted polar coordinate transformation and ALUB. As seen in Table VI, CMAKD's performance, when trained with limited labeled data, is comparable to the fully supervised SegFormer method. Furthermore, Fig. 8 visually demonstrates CMAKD's precision in segmenting MYO and LV regions, reaffirming its excellent generalization.

C. Impact of Additional Unlabeled OCT Data

Table VII shows the performance changes brought by progressively increasing the amount of same-modality unlabeled OCT data, where the additional unlabeled OCT images are obtained from [62]. For both SD-LayerNet and DiffTCL, adding unlabeled OCT data leads to consistent improvements,

but the gains gradually saturate as the data scale increases. When 1,000 unlabeled OCT images are introduced, the Dice scores improve by only 0.31 and 0.45, respectively. In contrast, when the amount of labeled CFP data is increased from 1,000 to 2,000, CMAKD achieves a Dice improvement of 1.59, indicating that the proposed cross-modality CFP assistance is more effective than simply increasing same-modality unlabeled OCT data. This also suggests that the proposed cross-modality CFP assistance provides more informative structural priors for OCT layer segmentation.

D. Limitations

Although the REFUGE2 dataset is derived from a glaucoma cohort, it does not provide detailed labels for refractive errors, axial length, or optic disc anomalies (such as tilted discs). Therefore, the generalization ability of the OCT segmentation method proposed in this study for eyes with high myopia or optic disc abnormalities needs further validation.

Furthermore, although the proposed CMAKD framework is theoretically capable of segmenting all retinal layers, it is currently constrained by the source domain model, which defines only two target categories. Retinal anatomy may vary across individuals due to factors such as axial length, high myopia, or optic disc anomalies (e.g., tilted discs), which may introduce structural distortions in OCT images. Since the public datasets used in this study do not provide metadata for these conditions, evaluating CMAKD under such clinical variations remains an important direction for future work. Consequently, our evaluation is restricted to the GC IPL and RNFL, as these two layers are more directly related to the structural information observable in CFP. Another limitation is that the structural priors transferred from CFP are not depth-resolved. Therefore, the proposed cross-modality guidance is biologically most interpretable for superficial OCT layers that are closely related to the optic nerve head, such as the RNFL and GC IPL. For deeper retinal layers, the benefit of CMAKD should be interpreted as indirect structural regularization rather than explicit transfer of layer-specific anatomy, and not as a substitute for OCT layer annotations. Therefore, although CMAKD shows promising performance on superficial structures, its robustness and clinical utility for comprehensive full-retina layer segmentation, especially for deeper or thinner structures such as the OPL or RPE, remain to be systematically validated. However, a systematic quantitative evaluation of other retinal layers and their potential applications in other retinal and choroidal diseases (e.g., Diabetic Macular Edema, Age-related Macular Degeneration) remain to be explored in future work.

VI. CONCLUSION AND FUTURE WORK

In this study, we addressed the challenge of limited annotated data in deep learning-based retinal layer segmentation from OCT images by proposing a novel CMAKD framework. By fully leveraging richly annotated CFP data as the source domain and transferring its shared structural priors to regularize the OCT model, together with a small set of annotated OCT images from the target domain, CMAKD effectively

achieves cross-modality feature transfer from CFP to OCT, thereby significantly improving the performance of retinal layer segmentation in OCT images. Specifically, we propose the DoCL module, which is designed to learn domain-invariant features, effectively narrowing the modality gap between CFP and OCT and promoting cross-modality feature alignment. In addition, we design the LA-MT framework to make full use of unlabeled OCT images and enhance the model's ability to capture subtle linear boundary features within the layered retinal structure, further improving segmentation accuracy. Extensive experiments conducted on three public datasets — GOALS, Duke DME, and MS — demonstrate that CMAKD achieves state-of-the-art performance, even under limited OCT annotation conditions. These results strongly validate the effectiveness of CMAKD in addressing cross-modality domain gap challenges and highlight its potential for clinical retinal OCT image analysis in real-world scenarios where annotated data is scarce. It should be noted that the additional modules in CMAKD are only used during training, while the inference stage relies solely on the segmentation backbone, resulting in a computational cost comparable to conventional segmentation models. In future work, we plan to extend CMAKD to other medical imaging modalities and explore the integration of lightweight network architectures, such as UniMatch [63], to further improve the scalability and deployment efficiency of our framework. Finally, extending this approach to fully unsupervised settings remains a challenging yet promising direction for our future research. In this study, the experimental evaluation focuses on the RNFL and GCIPL layers due to their clinical importance in glaucoma diagnosis and the availability of reliable annotations in the datasets used. Although the CMAKD framework is not inherently restricted to these layers, evaluating the method on full retinal layer segmentation requires datasets with consistent multi-layer annotations across domains. Extending the framework to comprehensive retinal layer segmentation will be explored in future work.

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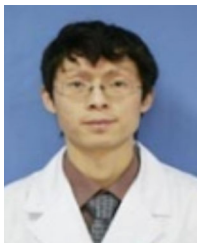
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